NR2DT-03: Step 3 — Design

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 3 of 10 | **Created:** April 2026 | **Last Updated:** 04/14/2026

Overview

Goal of this step: lock in the target Dynatrace architecture before any artifact migrates. Bucket strategy, host group taxonomy, IAM model, and OpenPipeline routing are all decided here.

Procedural runbook — see **USFOODS-G.01** (host groups) and **USFOODS-G.02** (Grail buckets) for the recommended design patterns this step uses.

Table of Contents

Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-02 — Strategize
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	USFOODS-G.01 / G.02 (governance recommendations)

1. What You'll Produce

Artifact	Format	Purpose
`bucket-design.md`	Markdown	Bucket names, retention, pricing model
`host-group-design.md`	Markdown	Host group taxonomy and naming
`iam-model.md`	Markdown	Groups, policies, scopes
`openpipeline-design.md`	Markdown	Enrichment + routing + filter rules
`terraform/`	HCL	Phase 1 Terraform module skeleton

2. Bucket Design

Use the recommended pattern from **USFOODS-G.02 Grail Buckets** as your starting template:

Bucket	Retention	Pricing	Purpose
`<org>_app_logs`	30 days	Retain w/ Included Queries	Application logs
`<org>_infra_logs`	14 days	Retain w/ Included Queries	Infrastructure / K8s
`<org>_security_logs`	365 days	Usage-based	Audit trail / PCI
`<org>_spans`	14 days	Retain w/ Included Queries	Distributed traces
`<org>_events`	90 days	Retain w/ Included Queries	Platform events
`<org>_bizevents`	90 days	Retain w/ Included Queries	Business events

Use `<org>_` as the prefix — it lets IAM policies scope with one `STARTSWITH "<org>_"` condition.

Document any deviations from this template (e.g., per-app buckets) and why.

3. Host Group Design

Per **USFOODS-G.01**, group hosts by ownership boundary, not by team alone. Common dimensions:

- Environment (`prod-app` , `nonprod-app` , `prod-infra`)
- Compliance scope (`payment-pci`)
- Workload tier (`prod-batch` , `prod-stream`)

Plan a host group per (environment × ownership) combination. Avoid a single all-hosts group — it breaks per-environment thresholds, IAM scoping, and automation safety.

4. IAM Model

Recommended starting groups:

Group	Bucket Access	Policy Condition
`<org>-admins`	All	`STARTSWITH "<org>_"`
`<org>-sre`	All operational	`bucket-name IN ('<org>_app_logs', '<org>_infra_logs', '<org>_events', '<org>_spans')`
`<org>-app-eng`	App logs only	`bucket-name == '<org>_app_logs'`
`<org>-readonly`	All except security	`STARTSWITH "<org>_" AND bucket-name != '<org>_security_logs'`

Always include an explicit DENY on `<org>_security_logs` for non-admin / non-audit groups.

5. OpenPipeline Layout

Routing rules in priority order (first match wins):

Priority	Matcher	Bucket
1	`matchesValue(log.source, "*audit*") OR loglevel == "CRITICAL"`	`<org>_security_logs`
2	`in(k8s.namespace.name, {"kube-system", "kube-public", ...})`	`<org>_infra_logs`
3	App namespace match	`<org>_app_logs`
4	catch-all	`default_logs` (review weekly)

Enrichment rules for any NR workload identifier (replaces NR's tag/workload model):

```
matcher: contains(k8s.namespace.name, "prod")
field: environment
value: prod
```

Drop rules for cost optimization (decide which apply now vs. Wave 5):

- Drop health-check logs (`contains(content, "GET /health")`)
- Drop DEBUG logs in production
- Sample sidecar proxy logs

6. Step Exit Criteria

G3 — Architecture Locked

- [ ] `bucket-design.md` final; bucket names + retention + pricing model documented
- [ ] `host-group-design.md` final; taxonomy understood by SRE + service owners
- [ ] `iam-model.md` final; security stakeholder signed off
- [ ] `openpipeline-design.md` final; routing rules sequenced
- [ ] Phase 1 Terraform module skeleton committed (will be applied in Wave 0 of Step 5)

Next step: NR2DT-04 — Translate (NRQL→DQL run; manual triage).

This notebook was AI-generated from community-submitted and publicly available sources, including the open-source Dynatrace-NewRelic, nrql-engine, and nrql-translator projects. This notebook series is not officially supported by Dynatrace or New Relic. Always verify information against the official Dynatrace documentation and New Relic documentation.